

Heart Attack Risk Prediction Using Voting Classifier Model based on Medical and Lifestyle Factors

Sarmiladevi G¹

Department of Mathematics, Amrita Vishwa Vidyapeetham, Coimbatore, India

Dheeraj P¹

Department of Mathematics, Amrita Vishwa Vidyapeetham, Coimbatore, India

Shubha Harini R V¹

Department of Mathematics, Amrita Vishwa Vidyapeetham, Coimbatore, India

Vibhushana Baskar¹

Department of Mathematics, Amrita Vishwa Vidyapeetham, Coimbatore, India

Vandhana S²

*Department of Computer Science and Engineering,
Amrita School of Computing, Coimbatore,
Amrita Vishwa Vidyapeetham, India
s_vandhana@cb.amrita.edu*

Abstract—Heart Attack is a life-threatening issue that is extremely dangerous. Early detection and treatment can reduce the mortality rates. It is estimated that youths are also prone to heart attacks no matter what their age is. Over the past years, many machine learning models were used to predict the risk of heart attacks across the world. This paper uses machine learning models to detect whether a person will have a heart attack or not with the help of hyper parameter tuning and voting technique. This study helps to know the possibility of occurrences of heart attack and can also help in reducing the future consequences by taking proper prior measures.

Index Terms—Heart Attack, machine learning, hyperparameter testing, probability, algorithm, chronic disease

I. INTRODUCTION

Millions of heart attacks are reported each year, making them one of the major causes of death globally. [1] Environmental factors, genetic predispositions, and changes in modern living are all closely associated with the growth of cardiovascular diseases-CVDs. Heart attacks continue to be a major public health concern in most of the countries. They can afflict people of all ages, geographical locations, and socioeconomic backgrounds. [2] Although traditional risk assessment models like ECG scans and blood test have proven useful in identifying individuals who are at risk, they sometimes overlook the numerous influencing factors like smoking, alcohol consumption, stress levels, and physical activity that are specific to each patient. One of the most significant challenges in predicting heart attacks is the variability of risk factors among individuals. The diversity in factors such as age, gender, region-urban vs. rural and lifestyle habits also play a crucial role. Another major challenge is handling large and complex medical datasets, where missing data, inconsistencies and biased samples can affect prediction accuracy. Addressing these challenges requires advanced analytical tools capable of recognizing

intricate patterns across multiple variables.

Because cardiac disease is complicated and multivariate, machine learning approaches present a promising method for predicting risk. ML algorithms can analyze enormous datasets and reveal hidden connections between variables in contrast to traditional models that depend on predetermined thresholds like [3] Thalassemia, Type of Chest-pain, ST depression, ECG while resting etc. By leveraging ML, we can develop a more dynamic and personalized prediction system that considers both medical and lifestyle-related factors. We can enhance early detection, support preventative care initiatives, and eventually lower heart attack death rates by using machine learning in predictive healthcare systems.

This research aims to develop a machine learning-based model for predicting heart attack risk using real-world patient data. The primary objectives includes the following:

- 1) Identify critical contributors to heart attack risk using variables such as age, gender, region, blood pressure, cholesterol, BMI, smoking, alcohol consumption, exercise level, sleep duration, stress level, and family medical history.
- 2) Apply multiple ML algorithms—Decision Tree, Balanced Random Forest, XGBoost, Adaptive Boosting, Gradient Boosting, Categorical Boosting, Gaussian Naive Bayes and Bagging. Each model was trained using different train-test splits and optimized using Grid-SearchCV to find the best hyper-parameters.
- 3) Evaluate all models with their accuracy, precision, recall, F1-score and obtain the best metric for each model to assess their predictive strength.
- 4) Combine the top-performing tuned models into a soft Voting Classifier to produce a more accurate and robust prediction system. This ensemble approach generates both classification outcomes and the probability% of

heart attack occurrence and non-occurrence.

This study contributes to the improvement of AI-driven medical diagnostics, aiding both clinicians and patients in mitigating the risks of heart disease before critical health events occur.

II. LITERATURE SURVEY

[4-7]”In recent years, applications of artificial intelligence technology, especially machine learning, in the field of auxiliary diagnosis have developed rapidly, and efficient progress has been made in automatic detection applications.” It includes platelet detection using improved YOLOv3, colon disease diagnosis through convolutional neural networks combined with optimization algorithms, and retinal optic disc localization leveraging swarm intelligence techniques. To create a machine learning model for predicting heart disease, data is collected from relevant medical records, including patient demographics, symptoms, and diagnostic test results. D. Singh. et al [8] applied ”various machine algorithms like Decision tree, PCA, SVM and Random forest. Among the algorithms used the random forest gives the best accuracy of 97%”.

To create an ML model for predicting the risk of heart disease, Karthick K. et al. [9] employed the SVM, Gaussian Naive Bayes, LR, Light GBM, XG Boost, and RF algorithms. The authors used feature selection to chose the best characteristics from the dataset on heart disease using Chi-square statistical test. It was observed that RF classifier had the highest accuracy rate of 88.5%. The cardiovascular disease rate in Singareni coal mining in Andhra Pradesh state was examined by K. Srinivas et al[10]. Data mining methods including Neural Networks, Decision Trees, and Naive Bayes have been employed. The Bayesian model obtained 82% classification accuracy with 87% sensitivity, whereas the decision trees-C4.5 achieved 82.5% classification accuracy with 87.17% sensitivity. ML obtained a 90.17% sensitivity and an 89.7% classification accuracy.

Mr. Santhana Krishnan.J et al[11] have predicted heart disease for Male Patient using Classification Techniques using Decision Tree with an accuracy of 91%. In Avinash Golande et al. [12] study of Heart disease Prediction Using Effective Machine Learning Techniques, suggested that experts use a few data mining techniques that are now accessible to help authorities or medical professionals identify heart illness. KNN classifier achieved the highest accuracy of 96%. According to G. Singh et al[13], the Decision Tree model outperformed all other ML and DL models in terms of accuracy, with 98.04%, nevertheless, MLP was also shown to be the second-best model, with 95.51% for binary heart disease categorization. the accuracy of logistic regression was obtained as 84.48% and accuracy of Naive Bayes was obtained as 83.12%.

V. Selvakumar et al[14] explored various machine learning models for heart attack prediction and reported that Logistic Regression achieved an accuracy of 91.2%, while Neural Networks, Bagging, and SVM showed competitive performance at 85.9%, 84.2%, and 84.1%, respectively. C.S Apte et al [15] prediction of heart disease was conducted, smoking and obesity were identified as key factors. The results showed that the ANN classification technique had the highest accuracy comparing to DT and NB on the used dataset.

Malavika et al[16] did an investigation on the usage of ML algorithms for predicting heart disease in people. Various ML algorithms like SVM, DT, NB, RF and KNN were implemented. From all the algorithms used, RF had the highest accuracy of 91.80% and Gaussian NB and SVM had the accuracy of 88.52%. From this, the decision that ML algorithms could be a useful tool in predicting heart disease and would aid in doctor’s diagnosis of the patients in a better way. Sahoo G. K. et al [17] made a comparison on the performances of LR, KNN, SVM, NB, DT, RF and XGBoost machine learning models for predicting heart disease for a heart disease dataset from the UCI ML repository. Among all algorithms used, RF gave the highest accuracy of 90.16%.

III. DATASET DESCRIPTION

This machine learning workflow begins from the process of data collection to model deployment. It includes data pre-processing, visualization, splitting, model development, evaluation, comparison, hyperparameter tuning, classifying using voting techniques and deployment, ensuring an optimized and efficient machine learning pipeline for accurate predictions.

A. Data Collection:

This dataset from Kaggle which contains 50,000 records related to heart health among youth and adults in Russia. It includes 23 features covering demographic details, lifestyle, and medical history, which may influence heart disease risk.

- Demographic information: Age, gender, region
- Health and medical history attributes: Blood pressure, cholesterol level, BMI, heart rate, diabetes, and family history.
- Lifestyle factors: Exercise habits, smoking, alcohol consumption, diet, sleep duration, and daily water intake.
- Medical history: Previous occurrences of heart attacks, use of medication, etc.
- Psychological factors: Stress levels, mental health status and health awareness.

B. Feature Analysis:

- Feature Correlation Analysis: To analyze numerical variables relationships, a correlation Heatmap(Fig. 1) was used and weak correlations indicates there’s a minimal or very loose relationship between the variables, and one variable’s change doesn’t strongly influence the other.

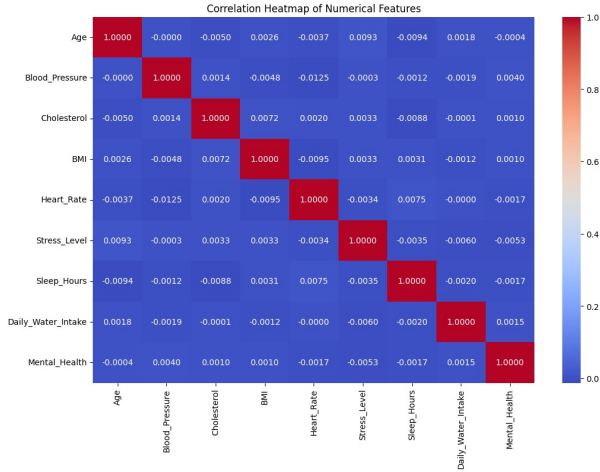


Fig. 1. Correlation Heatmap of Health Metrics

- Irrelevant features with no impact on the outcome were eliminated such as ID, Occupation, Income level, Education level, Marital status. Ensuring the model focuses only on meaningful features

C. Data Pre-processing:

Before deploying the model we perform the following preprocessing techniques

- Encoding Categorical Variables: For classification models, encoding was chosen based on efficiency. One-hot encoding was avoided due to high dimensionality. Instead, Label Encoding was applied to categorical variables, as tree-based models like Decision Trees, Random Forest, Gradient Boost and XGBoost effectively handle numerical representations of categories. This approach ensures optimal performance without introducing unnecessary complexity.
- Managing Missing Data: Effectively managing missing data is an essential step in ensuring the accuracy and reliability of data analysis. Initially alcohol consumption had 25024 null values. The null values in the numerical columns have been replaced with the mean value of the corresponding column values.

IV. METHODOLOGY

A. Data Splitting

Various train-test splits, including 70-30, 75-25, 80-20, and 85-15, were tested to assess the model's learning ability and generalization performance. Among these, an 80-20 or 70-30 split is commonly recommended in standard machine learning practices to maintain a balanced evaluation between training and validation. Nevertheless, following extensive testing and hyperparameter adjustment, the 75-25 split offered the best accuracy for ensemble models, which were further hyper-tuned for better prediction accuracy and stability.

B. Model Development

- 1) **Gaussian Naive Bayes:** Gaussian Naive Bayes is a type of classification algorithm which is based on Bayes theorem and follows normal distribution. It works with independent features only formula of Gaussian Naive Bayes is as follows:

$$P(Y|X) = \frac{P(X|Y) \cdot P(Y)}{P(X)} \quad (1)$$

Where:

- $P(Y|X)$ = Probability of class **Y** given features **X**
- $P(X|Y)$ = Likelihood of observing **X** given **Y**
- $P(Y)$ = Prior probability of class **Y**
- $P(X)$ = Probability of feature **X**

If a feature X_i follows a normal distribution, its probability is:

$$P(X_i|Y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(X_i - \mu)^2}{2\sigma^2}\right) \quad (2)$$

Here, μ (mean) and σ^2 (variance) are estimated from training data. The algorithm calculates probabilities for all classes and assigns the class with the highest probability.

- 2) **Balanced Random Forest:**

Balanced Random Forest addresses class imbalance by creating several decision trees. By resampling data, decreasing overfitting, and identifying intricate patterns, it improves accuracy and stability and is hence dependable for predicting heart attacks in medical diagnostics.

- 3) **Decision Tree:**

Recursively dividing the data into smaller subsets according to feature values is how decision trees work. This produces a structure that resembles a tree, with each node standing for a decision rule that produces a categorization result. A Decision Tree is used to identify patients as high or low risk for heart attacks based on criteria like age, cholesterol, and lifestyle choices.

- 4) **Adaptive Boosting:**

AdaBoost is a boosting method that increases the number of misclassified data points in order to gradually enhance model performance. Adjusted weights are used to fix past mistakes. AdaBoost effectively combines several weak classifiers, like decision stumps, to produce a powerful predictive model for heart attack prediction.

- 5) **Gradient Boosting:**

Another ensemble learning method is gradient boosting, which develops models in a stepwise manner and uses gradient descent to optimize each new learner to reduce

residual errors. Gradient Boosting is a very useful tool for medical risk assessment because it iteratively improves the accuracy of heart attack predictions.

6) **XG Boost:**

XGBoost, an optimized variant of gradient boosting, uses sophisticated strategies such as regularization and memory efficiency. XGBoost is a high-performance medical models since it can handle big datasets and intricate feature interactions in heart attack prediction.

7) **Categorical Boosting :**

CatBoost was created especially to manage categorical information effectively without requiring a lot of preprocessing. It maximizes computing efficiency and reduces overfitting. CatBoost is capable of accurately predicting heart attacks by analyzing patient information, including categorical variables like gender, smoking habits, and pre-existing diseases.

8) **Bagging:**

Bagging is an ensemble technique that improves model accuracy by training multiple models on different subsets of the data and combining their predictions. This approach helps reduce variance and enhances the reliability of predictions. By integrating the outputs of classifiers such as decision trees, bagging enhances model stability and reduces variance, making it effective for heart attack prediction.

C. Performance metrics

1) **Accuracy:**

Its used for calculating the number of correctly classified to the total number of instances:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

TP = True Positives (correctly predicted true cases)

TN = True Negatives (correctly predicted false cases)

FP = False Positives (incorrectly predicted true cases)

FN = False Negatives (incorrectly predicted true cases)

2) **Precision:**

Measure of predicted positive cases to actual positive cases:

$$Precision = \frac{TP}{TP + FP}$$

Higher precision implies that lesser incorrectly predicted True cases.

3) **Recall:**

Measures total number of positive cases that were correctly predicted:

$$Recall = \frac{TP}{TP + FN}$$

Higher recall means fewer incorrectly predicted false case.

4) **F1-Score:**

Measure of the harmonic mean of precision and recall, highly significant when there is a class imbalance:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

V. RESULT ANALYSIS

1) **Gaussian Naive Bayes:**

The performance of the Gaussian Naïve Bayes algorithm for heart attack prediction was evaluated across different train-test splits. The results show that the model achieved a consistent **88.24%** accuracy, 77.86% precision and 88.24% recall, resulting in an F1-score of 82.73% for all the splits.

2) **Balanced Random Forest:**

Balanced Random Forest showed varying performance across different train-test splits. The highest accuracy was observed in the 70-30 split with an accuracy of **87.87%** with precision of 79.58%. The model's highest F1-score was 82.87%.

3) **Decision Tree:**

The Decision Tree model was tested on different train-test splits, among which 85-15 split obtained the highest accuracy of **79.68%**, 79.72% precision and 79.68% recall, with the highest F1-score being 79.70%.

4) **XG Boost:**

The XGBoost model was evaluated using different train-test split ratios. The best performance was observed with 85-15 split, achieving **83.96%** maximum accuracy, 79.59% precision, 83.96% recall, and 81.59% F1-score.

TABLE I
PERFORMANCE METRICS:

Model	Best-Split	Accuracy
Gaussian Naïve Bayes	All Splits	88.24%
Balanced Random Forest	70-30	87.87%
Decision Tree	85-15	79.68%
XGBoost	85-15	83.96%

5) **Adaptive Boosting:**

The AdaBoost model, evaluated with varying numbers of estimators, achieved its highest accuracy of **88.64%** at 500 estimators, along with 78.57% precision, 88.64% recall, and 83.30% F1-score. Increasing the number of estimators beyond 500 resulted in a slight decline in accuracy, likely due to overfitting.

6) Gradient Boosting:

The Gradient Boosting model, evaluated with different numbers of estimators, attained its highest accuracy of **88.64%** at 100 estimators, with 78.57% precision, 88.64% recall, and 83.30% F1-score. However, increasing the number of estimators beyond 100 led to a slight decline in accuracy, which can be attributed to overfitting.

7) Categorical Boosting :

The CatBoost model, evaluated across different iteration values, achieved its highest accuracy of **88.64%** at both **500** and **100** iterations. At these points, the model also recorded 78.57% precision, 88.64% recall and 83.30% F1-score. Interestingly, increasing the iterations to 1000 or reducing them to 50 resulted in a slight drop in performance, indicating that excessive or insufficient training may lead to marginal overfitting or underfitting, respectively.

8) Bagging:

The Bagging model showed optimal performance at **500** and **100** estimators, achieving an accuracy of **88.61%** and **88.60%**, respectively. Both settings yielded strong F1-scores of 83.29% and 83.30%, with precision and recall values reflecting consistent performance. Notably, the model attained its highest precision of 80.86% at 50 estimators, though with a slight decrease in overall accuracy. These results indicate that moderate estimator counts strike a balance between learning and generalization in the Bagging approach.

TABLE II
PERFORMANCE METRICS:

Model	Estimators	Accuracy
AdaBoost	500	88.64%
Gradient Boosting	100	88.64%
CatBoost	500	88.64%
Bagging	500	88.61%

TABLE III compares various machine learning models based on their optimal data split or the number of estimators used and their corresponding accuracy. Ensemble learning techniques such as Gradient Boosting, AdaBoost, CatBoost achieve the highest accuracy, approximately **88.64%**, indicating their strong predictive capabilities. Bagging, Gaussian Naïve Bayes and Balanced Random Forest also perform well, with accuracy rates of 88.61%, 88.24% and 87.87%, respectively. On the other hand, the Decision Tree model has the lowest accuracy of 79.68%. Models trained with an 80-20 or 85-15 data split other than Ensemble methods tend to perform better.

The proposed model employs a **Voting Classifier** (Fig.2) that integrates multiple high-performing machine learning algorithms, including Gaussian Naïve Bayes, Balanced Random Forest, Decision Tree, AdaBoost, Gradient Boosting,

TABLE III
PERFORMANCE METRICS FOR DIFFERENT SPLITS

Model	Best-Split	Accuracy
Gaussian Naïve Bayes	All Splits	88.24%
Balanced Random Forest	70-30 Split	87.87%
Decision Tree	85-15 Split	79.68%
AdaBoost	500 Estimators	88.64%
Gradient Boosting	100 Estimators	88.64%
XGBoost	85-15 Split	83.96%
CatBoost	500 Estimators	88.64%
Bagging	500 Estimators	88.61%

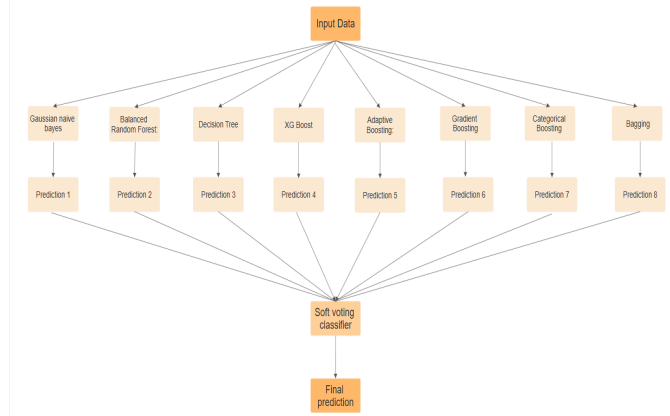


Fig. 2. Voting Classifier diagram

XGBoost, CatBoost, and Bagging. By leveraging soft voting, the model aggregates probabilistic predictions from these classifiers to enhance overall accuracy. The results demonstrate that this ensemble method improves classification performance by considering diverse learning paradigms. The evaluation metrics, including accuracy of **81.05%**, precision of 79.40%, recall of 81.05%, and F1-score of 80.20%. This model shows promising potential for robust heart disease prediction, enabling more reliable risk assessment.

VI. CONCLUSION

In order to predict cardiac disease, we investigated on a number of machine learning algorithms, evaluated their effectiveness, and chose the top models. We used a voting technique to choose the majority forecast from the top-performing models. This method made it possible to forecast a person's chances of getting a heart attack in a more consistent and accurate manner. The graph (Fig.3) indicates that while the total number of heart attacks is higher in youth 3,378 cases than in the aged 2,503 cases, the elderly have a greater relative risk. This highlights the importance of preventive measures for all age groups.

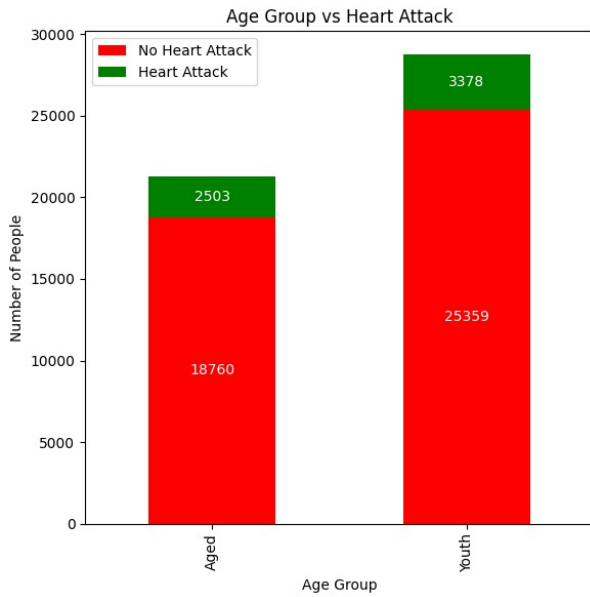


Fig. 3. Age Group vs Heart Attack

Similarly, we have trained these machine learning algorithms on heart disease data from France and India. The best metrics for each dataset have been tabulated below (Table IV and Table V). Future studies can concentrate on improving feature selection and incorporating deep learning for more accurate early identification and individualized risk assessment, even though our model shown encouraging accuracy.

TABLE IV
PERFORMANCE METRICS FOR FRANCE DATASET:

Model	Accuracy	Precision	Recall	F1 Score
CatBoost	79.93	63.88	79.93	71.01
AdaBoost	79.93	63.88	79.93	71.01
Gradient Boosting	79.93	63.88	79.93	71.01
Balanced Random Forest	79.90	67.00	79.90	71.00
Bagging	79.62	69.00	79.62	70.65
XGBoost	79.42	68.32	79.42	70.66
Gaussian Naïve Bayes	79.96	63.93	79.96	71.05
Decision Tree	67.84	68.11	67.84	67.97
Voting Classifier	69.90	68.00	69.90	69.00

TABLE V
PERFORMANCE METRICS FOR INDIA DATASET:

Model	Accuracy	Precision	Recall	F1 Score
CatBoost	80.48	71.41	80.48	72.00
AdaBoost	80.56	64.90	80.56	71.89
Gradient Boosting	80.52	64.89	80.52	71.87
Bagging	80.56	74.70	80.56	72.19
Balanced Random Forest	79.56	63.60	79.56	70.69
XGBoost	74.75	67.97	74.75	70.44
Gaussian Naïve Bayes	79.77	63.63	79.77	70.79
Decision Tree	68.73	68.43	68.73	68.58
Voting Classifier	70.87	68.00	71.00	69.00

VII. REFERENCES

- 1) Aruni Bhatnagar "Environmental Determinants of Cardiovascular Disease"
- 2) M. D. Saryazdi et al., "Identification and validation of key predictive factors for heart attack diagnosis using machine learning and fuzzy clustering."
- 3) Rahul Katarya¹ Sunit Kumar Meena¹ "Machine Learning Techniques for Heart Disease Prediction: A Comparative Study and Analysis"
- 4) R. Liu, C. Ren, M. Fu, Z. Chu, and J. Guo, "Platelet Detection Based on Improved YOLOv3," *Cyborg Bionic Systems*, vol. 2022, 9780569, 2022.
- 5) A. A. A. Mohamed, A. Haçerlioğlu, J. Rahebi, M. K. Ray, and S. Roy, "Colon Disease Diagnosis with Convolutional Neural Network and Grasshopper Optimization Algorithm," *Diagnostics*, vol. 13, p. 1728, 2023.
- 6) J. Rahebi, "Fishier Mantis Optimizer: A Swarm Intelligence Algorithm for Clustering Images of COVID-19 Pandemic," *International Journal of Nanotechnology*, vol. 20, pp. 25–49, 2023.
- 7) N. F. A. AlShalchi and J. Rahebi, "Human Retinal Optic Disc Detection with Grasshopper Optimization Algorithm," *Multimedia Tools and Applications*, vol. 81, pp. 24937–24955, 2022.
- 8) A. Singh, H. Mahapatra, A. K. Biswal, M. Mahapatra, D. Singh, and M. Samantaray, "Heart Disease Detection Using Machine Learning Models," in *Procedia Computer Science*, vol. 235, pp. 937–947, 2024, International Conference on Machine Learning and Data Engineering (ICMLDE 2023).
- 9) K. Karthick, S. K. Aruna, R. Samikannu, R. Kuppusamy, Y. Teekaraman, and A. R. Thelkar, "Implementation of a Heart Disease Risk Prediction Model Using Machine Learning," *Computational and Mathematical Methods in Medicine*, vol. 2022, p. 6517716, 2022.
- 10) K. Srinivas, G. R. Rao, and A. Govardhan, "Analysis of Coronary Heart Disease and Prediction of Heart Attack in Coal Mining Regions Using Data Mining Techniques," in *Proceedings of the 2010 5th International Conference on Computer Science and Education (ICCSE)*, IEEE, pp. 1344–1349, 2010.
- 11) S. Krishnan and S. Geetha, "Prediction of Heart Disease Using Machine Learning Algorithms," in *Proceedings of the 2019 1st International Conference on Innovations in Information and Communication Technology (ICIICT)*, doi:10.1109/ICIICT1.2019.8741465.
- 12) A. Golande and P. T. Kumar, "Heart Disease Prediction Using Effective Machine Learning Techniques," *International Journal of Recent Technology and Engineering (IJRTE)*, vol. 8, no. 1S4, June 2019.
- 13) G. Singh, K. Guleria, and S. Sharma, "Machine Learning and Deep Learning Models for Early Detection of Heart Disease," in *Proceedings of the 2023 International Conference on Computing, Communication, and Intelligent Systems (ICCCIS)*.

- 14) V. Selvakumar, A. Achanta, and N. Sreeram, "Machine Learning-Based Chronic Disease (Heart Attack) Prediction," in *Proceedings of the 2023 International Conference on Innovative Data Communication Technologies and Applications (ICIDCA)*, IEEE Xplore, ISBN: 979-8-3503-9720-8.
- 15) C. S. Apte, "Improved Study of Heart Disease Prediction System Using Data Mining Classification Techniques," *International Journal of Computer Applications*, vol. 47, pp. 44–48, 2012.
- 16) A. Singh and R. Kumar, "Heart Disease Prediction Using Machine Learning Algorithms," in *Proceedings of the 2020 International Conference on Electrical and Electronics Engineering (ICE3)*, Gorakhpur, India, 14–15 February 2020, pp. 452–457.
- 17) G. K. Sahoo, K. Kanike, S. K. Das, and P. Singh, "Machine Learning-Based Heart Disease Prediction: A Study for Home Personalized Care," in *Proceedings of the 2022 IEEE 32nd International Workshop on Machine Learning for Signal Processing (MLSP)*, Xi'an, China, 22–25 August 2022, pp. 1–6.